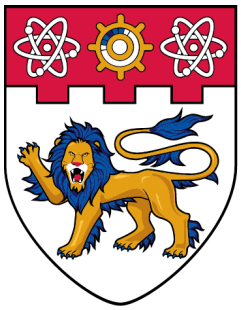
NTU BRIDGE DESIGN COMPETITION 2022



NANYANG TECHNOLOGICAL UNIVERSITY
SINGAPORE

Technical proposal

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Section 1: Proposed Bridge Design and Alignment

1.1 Proposed Bridge Design

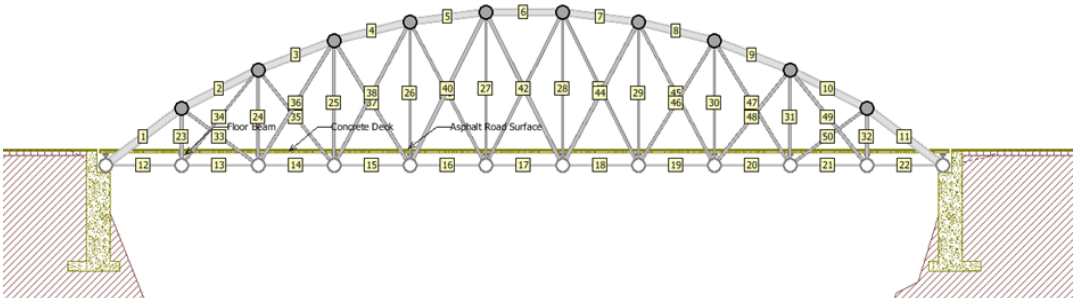


Figure 1: Proposed primary frame of the main arch bridge

The primary frame of the main arch bridge can be split into four different parts: arch frame, deck, tension cable, and truss. Members 1-11 form the arch frame, which is under compression force. Members 12-22 form the deck, which is under tension force. Members 22-32 are tension cables, which are under tension force. Lastly, members 33-50 are truss members, which are under tension and compression.

The secondary arch bridge is an arch bridge of similar design to the main arch bridge.

1.2 Slope Design of Road

The main arch bridge and secondary arch bridge are to be constructed at the elevation of 5 m and 1.2 m with the Pang Sua Park connector being the horizontal datum. According to LTA COP for street work proposals [1], the slope of a road should not exceed 8%.

To connect to the main bridge from the south connection point, which is at an elevation of 6m, flexible pavement roads are needed. A road of 30m with a slope of 3.33% is to be constructed. To ensure a 3m headroom for the Rail Corridor a road of 55m with a slope of 3.27% is needed, leaving 3.2m of headroom at the Rail Corridor. Continuously, a road of 32m with a slope of 6.25% is to be constructed, connecting to the secondary bridge.

The distance from the secondary bridge to the north connection point is 20m, therefore, a road of greater slope is needed. According to the LTA COP for street work proposals [1], when a road steeper than 1:30 approaches a junction, a minimum length of 10m with a gradient of not more than 1:50 needs to be constructed before meeting the road. Therefore, a road of 10m with a slope of 2% followed by a road of 10m with a slope of 8% is constructed to connect the secondary bridge to the north connection point. Due to the steepness of the road, the road is to be of rigid pavement.

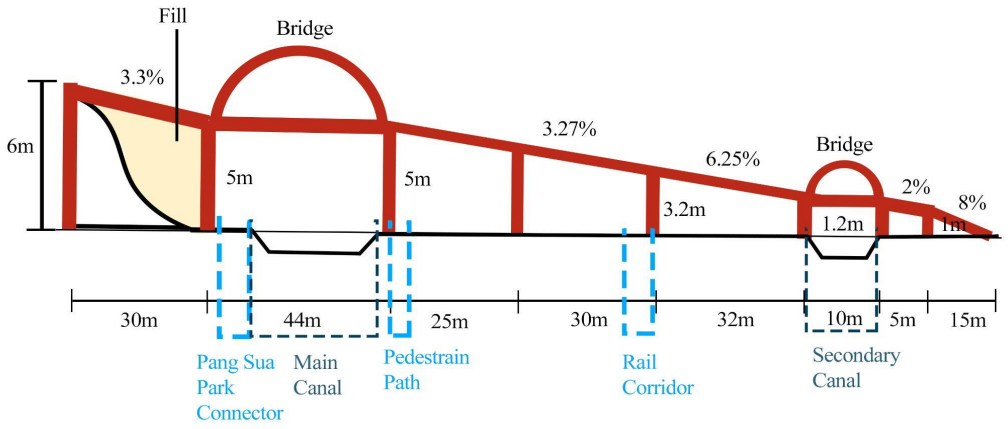


Figure 2: Slope Design of Road

Firstly, excavation for the piers is done, a retaining wall is constructed and landfilling is added to the natural slope. Then the piers and bridge abutments are constructed. Lastly, the roads are constructed.

Section 2: Method Statement and Construction Sequence

2.1 Summary of Construction Process

Construction Process	
Phase 1: Approval of Design and Proposal	
1	Approval for the design of slope, bridge and piers from Land Transport Authority (LTA) and Building & Construction Authority (BCA)
2	Analysis and reporting from arborist on green space, approval for transplant from National Parks (NParks).
3	Approval from stakeholders and nature and heritage groups.
Phase 2: Preliminary	
2.1 Setting Out	
1	Site investigation and soil analysis.
2	Mapping out gridlines and erecting site hoarding, marking contract boundaries and installation of site hoarding and demarcating service lines.
3	Vegetation removal and soil excavation to the foundation level's bottom. Installation of settlement monitoring devices.
4	Installation of acoustic hoarding barriers, noise monitoring devices, netting and shielding.

5	Placement of signs, certifications and relevant documents.
2.2 Off-site Preliminary works	
1	Prefabrication of concrete deck and steel deck.
Phase 3: Foundation and Abutments	
3.1 Pile Foundation	
1	Create void using the boring machine at the location of pile and pier (Figure 7).
2	With a casing placed in the void, cast steel-reinforced concrete.
3	Construct reinforcement pile cap.
3.2 Construction of Abutments and Piers[2]	
1	Fixing a grillage of steel reinforcements for piers and abutments.
2	Erection of vertical formwork for piers and abutments with shores and bracing.
3	Fill formwork with concrete and compact by vibrating.
4	Remove shutters, curing of concrete, treating exposed surfaces and application of waterproof membrane(bitumen coating applied by brush or spray) on faces retaining soil fill.
5	Laying of the drain(small pipe with granular surround) behind abutments.
6	Place and compact granular fill, behind abutments and wing walls, to road formation-level.
Phase 4: Construction of Bridge	
4.1 Arch Frame	
1	The arch frame is bolted and assembled on-site.
2	Using a crawler crane, lift and install bridge arch frame (Figure 9 in Appendix).
4.2 Bridge Deck	
1	Using a crane to place precast concrete beams atop temporary supports at abutments.
2	Install steel deck framing along floor beams.
3	Install the precast concrete deck on top of steel deck framing.
4	Above the precast concrete deck, finish with asphalt road surface.

2.2 Foundation and Soil Work

Crawler excavator or backhoe excavator is used, which is favorable on sites with uneven terrain. Deep foundation is chosen for this project. It is necessary where the bearing capacity of the surface soils is insufficient to support loads imposed and so they are transferred to deeper layers with a higher bearing capacity[3]. The soil condition may be poor and not stable enough as it is located near a canal and drainage system. Temporary casing is needed to protect the soil from collapsing into the borehole during drilling. Site investigation and soil analysis must be done.

Sheet pile or bored pile retaining wall is used. Bored pile is more suitable due to the vibration or noise levels generated by a pile driver are not acceptable due to the concerns of residents, students, teachers, the elderly, and nursing home staff. Earth Retaining Stabilizing Structure (ERSS) design is used to maintain the shape of excavation during construction and earth filling. [4]

2.3 Construction of Arch Frame

The length of the arch frame for the main bridge is 44m, for the secondary bridge is 15m. Vehicles longer than 13 meters are not permitted on LTA roads, hence the major frame span must be split and not exceed 12 meters. It is recommended that each member be sent to the site separately. Then there are the members of the primary school.

On-site, the frame is welded together. Instead of using screws, the members' connections are designed to be bolted on site. Welding, by its very nature, necessitates a high level of consistency and craftsmanship. Simultaneously, bolting has a reduced requirement for craftsmanship and has a faster assembly pace. [5]

2.4 Assembly of Truss members and Tension Cables

Gusset plates are fitted at the joining point of the arch frame components. Before the arch frame is erected, the truss members are joined to the arch frame on the gusset plates, and the tension cables are also attached to the arch frame's gusset plate on the floor. [6] The principal frame size (Figure 9 in Appendix) is located on the job site and is small enough to be erected on the spot.

2.5 Erection of Primary Frame and Equipment used

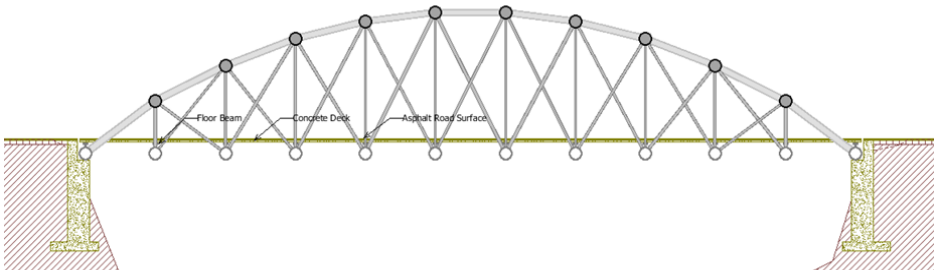


Figure 3: Primary frame without deck

Based on the software, the weight of the primary frame without decks (Figure 3) is 4.436 tons. A rented or second-hand crawler crane (model Kobelco 7070-1F [7]) with a maximum lifting capacity of 5 tons at 27.7 m working radii is used to lift the mainframe.

The crane is sufficient as the maximum weight of the arch frame, which is about 4.5 tons at 22 m, can carry the maximum load of 7.1 tons at this distance. Thus, the factor of safety is 1.6 (<1, safe range). The crane consists of an engine with a low noise level, suitable for the construction done near residential areas. The crane is installed on a tracked vehicle with metal tracks that increase the crawler crane's stability on the ground, suitable for all types of grounds and uneven terrain.

Temporary propping systems are installed at the base of the primary frame to ensure the primary frame's stability. The second frame is done the same.

2.6 Prefabricated Steel Deck Framing and Lateral Stability

The 4mx10m deck framings are constructed off-site (Figure 12 in Appendix). It's worth mentioning that specific road approvals are necessary when the items being transported have a width of more than 3m. The steel deck must be moved during off-peak hours(0700-0930 and 1700 to 2130), even if the LTA has granted authority. As a result, this must be accounted for in the construction schedule and planning.

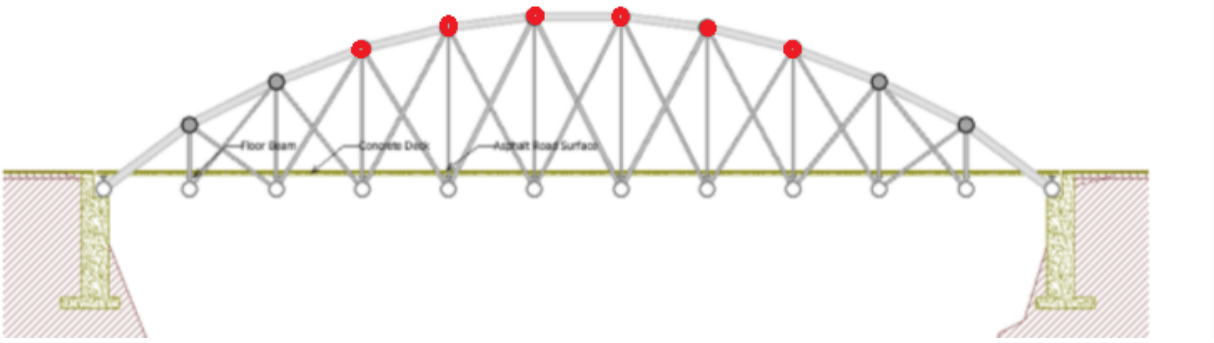


Figure 4: Location of lateral truss and bracing connection points (Red).

With the aid of the crawler crane, the steel deck framings may subsequently be raised and constructed on the project site. The installation technique is depicted in Figure 10. To connect the constructed primary frame to the framing, the gusset plate [6] is prefabricated on the steel deck framings. There's also the question of lateral stability to think about. Figure 4 shows the lateral truss and bracing connection positions in red. Throughout the prefabricated steel deck, bracing should be fitted. Its goal is to account for the bridge's lateral stability, particularly to reduce global buckling for the main arch and other horizontal loadings like wind or unintended vehicle loading. Finally, the tension wires must be tightened to achieve their strength before the concrete deck and asphalt road are installed. The steel deck framework may be built on top of the precast concrete

deck, and asphalt concrete can be poured on top of it. The superstructure of the bridge has been completed. [8]

Section 3: Site Constraints and Proposed Solutions

3.1 Traffic Management

To facilitate logistics and delivery, traffic management is required. The construction site can be accessed from both the south and north side of the construction site. From the south, the site is accessible from Chua Chu Kang North 6, a two-way road. From the north, the site is accessible from 363 Woodlands Rd, a one-way road.

The south and north connection point are both junctions, therefore, control measures are needed:

- Consult relevant authorities for traffic safety requirements for the locations
- Provide traffic signboards and stop signs where necessary
- Provide traffic controller as required

For the south construction site, traffic signboards and a traffic controller is required. Traffic controllers need to assist with vehicles entering and exiting the site, and also facilitate pedestrians crossing along the pedestrian walk which intersects with the entry of the site. For exit from the junction, a stop sign is placed.

On the north construction site, the road connecting to Woodlands Rd is to be modified to a two-way road. The road lines are to be repainted and the car parking spots are to be removed. To facilitate the entry and exit of vehicles at site entry and the crossing of pedestrians, a traffic controller is needed. Traffic signboards are to be placed at the road junction.

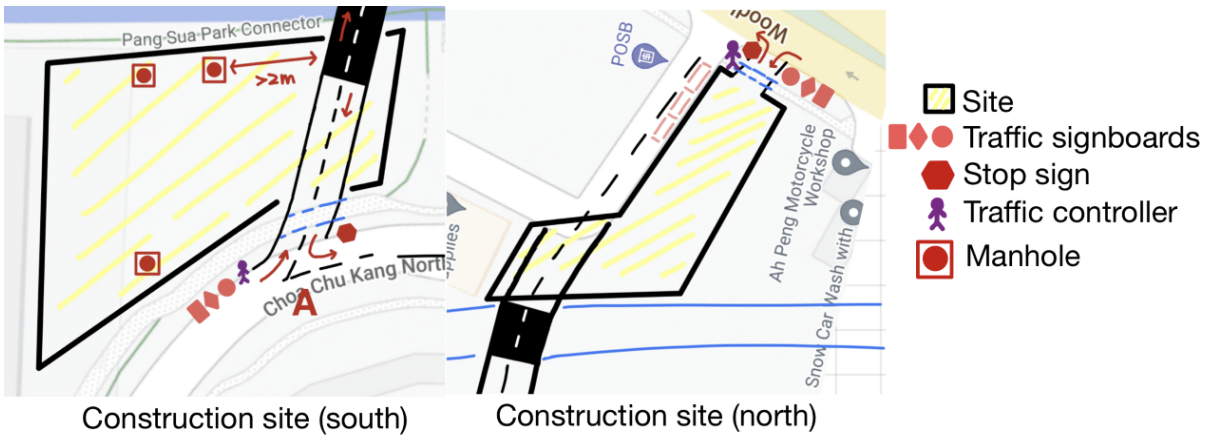


Figure 5: Traffic plan surrounding site

As crawler cranes are used, it is important to note that they are big and bulky vehicles, and are not suitable to be transported as it is. During construction, parts of the crawler crane are to be

transported separately to the site via trucks and to be assembled on site. Once the work is done, the crawler crane needs to be dismantled and transported out separately.

The entry and exit of site vehicles need to be guided by traffic controllers as they are big vehicles. The unloading of materials and assembly of the crawler crane is to be done only within the site boundary. The site is surrounded by steel plate barriers for safety purposes and to avoid disruption caused to nearby facilities.

A temporary route is created on the east side of the site. This is for the transportation of construction vehicles and resources to the site. Traffic signboards and a traffic controller are needed to guide the transportation process.

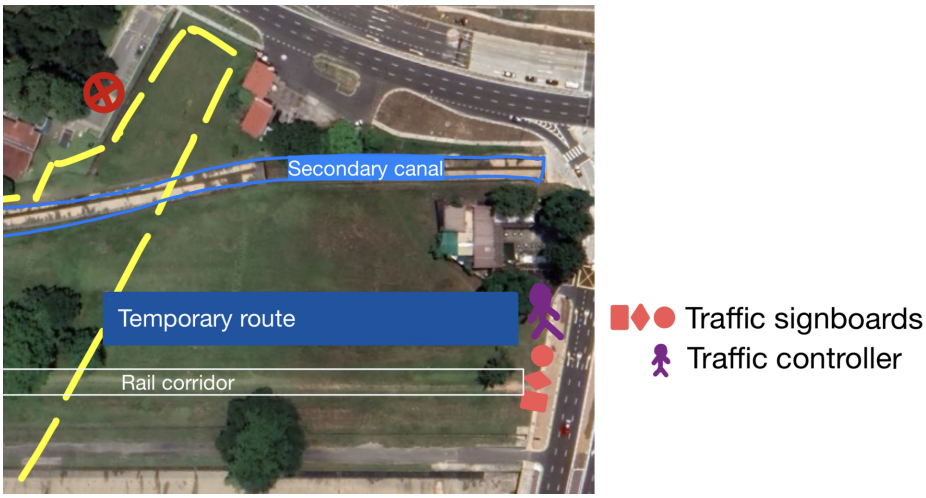


Figure 6: Location of temporary route

3.2 Site Procession - Hoarding Plan

Choa Chu Kang North 6 at River South now has a 10m wide access. At River North's Woodlands Road, a new 10m wide entry has been opened. Figure 7 depicts the site entrance, site office, and storage area.

There is a 10m clearance from the river to the South (Figure 13 in Appendix), allowing for a 4m wide and 3m clear headroom for the pedestrian path to remain accessible during all construction stages and the drainage setback.

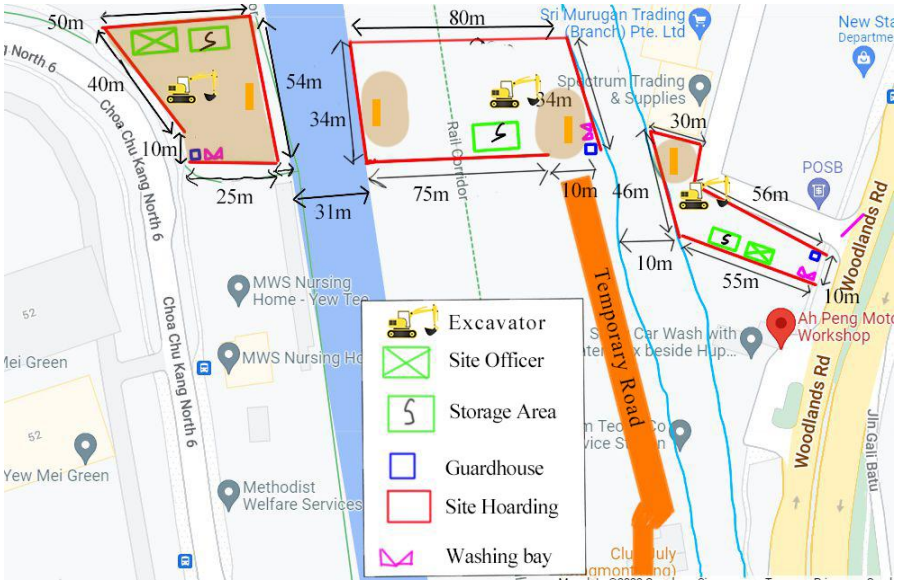


Figure 7: Proposed Site Plan

3.3 Public Sewerage System

There are three sewer manholes and a drainage canal at Project South. Based on the Protection of Public Sewerage System by Public Utilities Board (PUB) [9], sewer setback net clearance for a sewer diameter 150-600mm, depth>5m is 2.0m from the outer edge of any structure to the centreline of sewer pipe. Therefore, the roads connecting to the bridge must be constructed at least 2m away from the manhole and drainage (Figure 5). We are required to engage a PUB registered CCTV contractor to carry out pre-construction and post-construction inspections.

Section 4: Environmental impacts and measures

4.1 Trees in green space

Based on Figure 14 in Appendix, the trees will need to be transplanted to the land nearby for the bridge to be built. We will need to seek advice from an arborist for the procedures of transplanting the trees. Only after detailed analysis and approval by Land Transport Authority (LTA) & National Parks (NPARKS), will the construction proceed.

After the construction of the bridge, we propose planting more trees in the green spaces available. Shrub planting would be considered along the bridge (Figure 15 in Appendix).

4.2 Rail Corridor

The rail corridor is a green space and heritage area, hence the construction area will be planned to avoid the rail corridor. Fencing to enclose the area of the rail corridor under and near the piers (Figure 14). The planning of the construction process, roads, and position would take the rail

corridor into consideration. The rail corridor at this area would be closed to residents/tourists during the construction process.

4.3 Others

Materials:

- Reuse construction waste (from previous projects). For example, concrete waste can be reused as reclamation material. Isolation and identification of recyclable construction waste should be done in the beginning.
- Recyclable concrete, bricks could be used. For example, green concrete is used. It uses waste material as at least one of its components, or its production process does not lead to environmental destruction, or it has high performance and life cycle sustainability. [10]
- Design the construction considering overall cost and avoiding wastage in materials.

Equipment:

- Using hybrid equipment: Using hybrid equipment that runs on both fossil fuels and electricity helps cut down on fuel consumption and greenhouse gas emissions as well.
- Minimizing vehicle idling time: When vehicles idle, they continue to burn fossil fuels, even though they're not going anywhere.
- Alternative solutions like inflatable dams and soil netting are gentler on local ecosystems compared to sandbags. [11]

Section 5: Other Considerations

5.1 Site Safety

Several actions are considered to ensure site safety.

1. Toolbox Meeting
   It will be held every morning before commencing work to brief the workers on daily activity plans so that everyone on site is aware of the whole site's activities, hazards, and safe work practices.

2. Daily inspections
   PPE inspections, checking, and verification of the on-field compliance including Safety Standards, working conditions, and other critical areas of the worksite. Housekeeping programmes must be done, the checklist of tidiness and cleanliness on site is prepared.

3. COVID-19 safety measures
   Routine swab tests (Antigen Rapid Test) for all workers will be prepared, workers are required to be fully vaccinated (or otherwise required to do employer-supervised ART every 3 days). [12]

5.2 Soil Improvement

Strengthening the surrounding soil through grouting to the ground to avoid excess settlement for the piers and abutment. Grouting is relatively costly and so wastage must be controlled. The stability of slopes can be increased to good effect by anchoring them into the bedrock below the probable shear surface.[13] This is to ensure that heavy machinery can be placed on top of the slope, e.g., excavator and crane.

At Project North, there are many low-rise buildings (1-2 storeys) serving as workshops and shophouses. There are also similar buildings at Project South. These buildings are likely to sit on shallow foundations, which are sensitive to settlement. Strengthening of the surrounding soil or retaining wall is needed, and a settlement monitoring device is required to track the settlement of the soil before and throughout the construction process.

5.3 Noise and Dust Control

At Project South, there is a primary school, nursing home, and housing area. Noise cancellation methods like acoustic hoarding barriers are installed before construction works to minimize the disruption to the surrounding residents and also to the biodiversity in the green space, reducing noise pollution. As mentioned in Section 2.2, bored piles are used rather than driven piles to minimize vibrations and soil movements, minimizing noise pollution and the influence on adjacent buildings. Restriction of noisier activities like piling, concreting, and demolition to day-time (after 3 pm). Besides, noise monitoring devices and vibration monitoring devices are also used to ensure both limits are maintained throughout the whole construction, based on Table 1.

For concerns of dust pollution, effective measures such as water sprinklers/spray, shielding, netting, covers/hoarding for aggregate and sand storage would be taken to minimize dust pollution caused by construction works. The netting or barriers on the scaffolding of the construction site shall be of suitable height for effective containment of dust and debris. All construction debris should be properly stored and removed for disposal quickly. They should not be left to accumulate at the site.[14]

Boundary noise limits for factory premises

Types of affected premises	Maximum permitted noise levels (reckoned as the equivalent noise level over the specified period) in decibels (dBA)		
	Day (7am-7pm)	Evening (7pm-11pm)	Night (11pm-7am)
Noise Sensitive Premises	60	55	50
Residential Premises	65	60	55
Commercial Premises	70	65	60

Table 1[15]

Appendix



Figure 8: Construction sequence of road

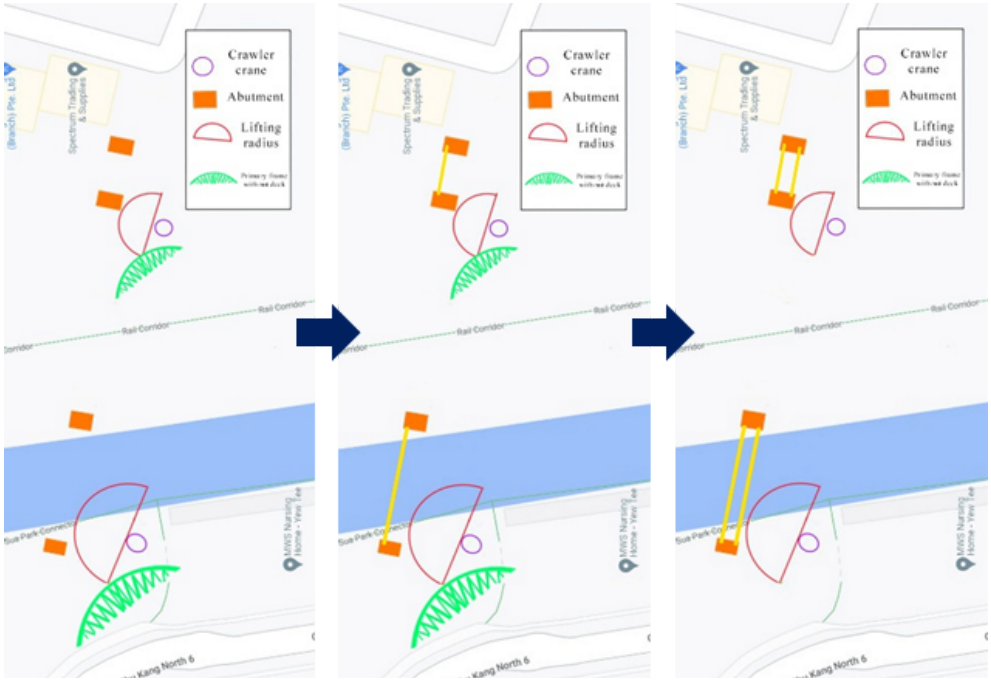


Figure 9: Assembly of Primary frame

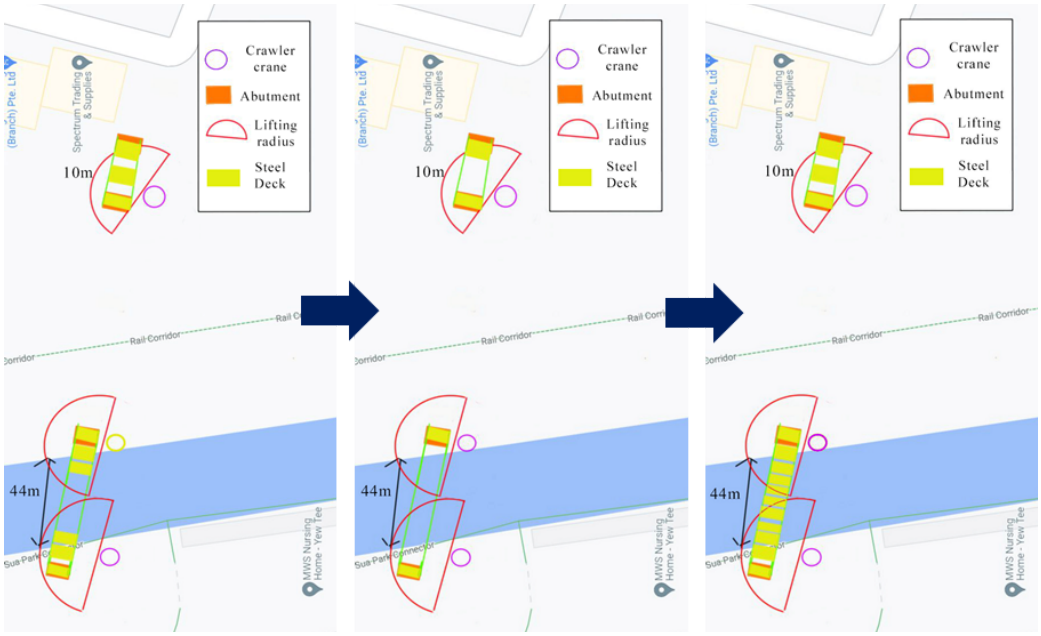
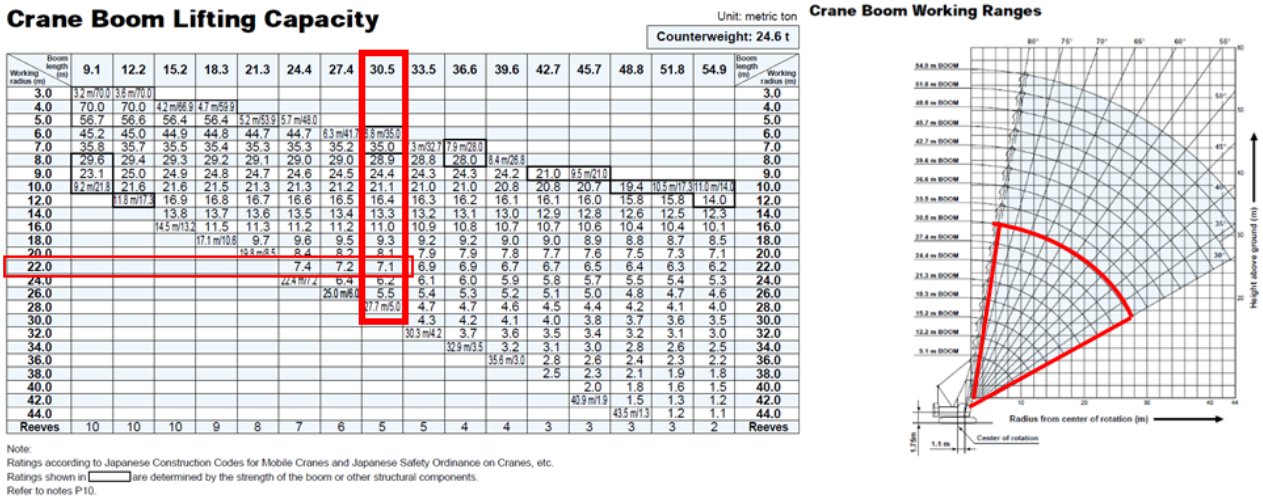


Figure 10: Installation of steel deck



Crane Boom Lifting Capacity

Unit: metric ton

Counterweight: 24.6 t

Working radius (m) \ Boom length (m)	9.1	12.2	15.2	18.3	21.3	24.4	27.4	30.5	33.5	36.6	39.6	42.7	45.7	48.8	51.8	54.9	Boom length (m) \ Working radius (m)
3.0	3.2 m/70.0	3.6 m/70.0															3.0
4.0	70.0	70.0	4.2 m/66.9	4.7 m/59.9													4.0
5.0	56.7	56.6	56.4	56.4	5.2 m/53.9	5.7 m/48.0											5.0
6.0	45.2	45.0	44.9	44.8	44.7	44.7	6.3 m/41.7	6.8 m/35.0									6.0
7.0	35.8	35.7	35.5	35.4	35.3	35.3	35.2	35.0	3 m/32.7	7.9 m/28.0							7.0
8.0	29.6	29.4	29.3	29.2	29.1	29.0	29.0	28.9	28.8	28.0	8.4 m/26.8						8.0
9.0	23.1	25.0	24.9	24.8	24.7	24.6	24.5	24.4	24.3	24.3	24.2	21.0	9.5 m/21.0				9.0
10.0	9.2 m/21.8	21.6	21.6	21.5	21.3	21.3	21.2	21.1	21.0	21.0	20.8	20.8	20.7	19.4	10.5 m/17.3	11.0 m/14.0	10.0
12.0		11.8 m/17.3	16.9	16.8	16.7	16.6	16.5	16.4	16.3	16.2	16.1	16.1	16.0	15.8	15.8	14.0	12.0
14.0			13.8	13.7	13.6	13.5	13.4	13.3	13.2	13.1	13.0	12.9	12.8	12.6	12.5	12.3	14.0
16.0			14.5 m/13.2	11.5	11.3	11.2	11.2	11.0	10.9	10.8	10.7	10.7	10.6	10.4	10.4	10.1	16.0
18.0				17.1 m/10.6	9.7	9.6	9.5	9.3	9.2	9.2	9.0	9.0	8.9	8.8	8.7	8.5	18.0
20.0					19.8 m/8.5	8.4	8.2	8.1	7.9	7.9	7.8	7.7	7.6	7.5	7.3	7.1	20.0
22.0						7.4	7.2	7.1	6.9	6.9	6.7	6.7	6.5	6.4	6.3	6.2	22.0
24.0						22.4 m/7.2	6.4	6.2	6.1	6.0	5.9	5.8	5.7	5.5	5.4	5.3	24.0
26.0							25.0 m/6.0	5.5	5.4	5.3	5.2	5.1	5.0	4.8	4.7	4.6	26.0
28.0								27.7 m/5.0	4.7	4.7	4.6	4.5	4.4	4.2	4.1	4.0	28.0
30.0									4.3	4.2	4.1	4.0	3.8	3.7	3.6	3.5	30.0
32.0									30.3 m/4.2	3.7	3.6	3.5	3.4	3.2	3.1	3.0	32.0
34.0										32.9 m/3.5	3.2	3.1	3.0	2.8	2.6	2.5	34.0
36.0											35.6 m/3.0	2.8	2.6	2.4	2.3	2.2	36.0
38.0												2.5	2.3	2.1	1.9	1.8	38.0
40.0													2.0	1.8	1.6	1.5	40.0
42.0													40.9 m/1.9	1.5	1.3	1.2	42.0
44.0														43.5 m/1.3	1.2	1.1	44.0
Reeves	10	10	10	9	8	7	6	5	5	4	4	3	3	3	3	2	Reeves

Note:
Ratings according to Japanese Construction Codes for Mobile Cranes and Japanese Safety Ordinance on Cranes, etc.
Ratings shown in ☐ are determined by the strength of the boom or other structural components.
Refer to notes P.10.

Crane Boom Working Ranges

Figure 11: Crane Boom Lifting Capacity and Working Ranges [7]

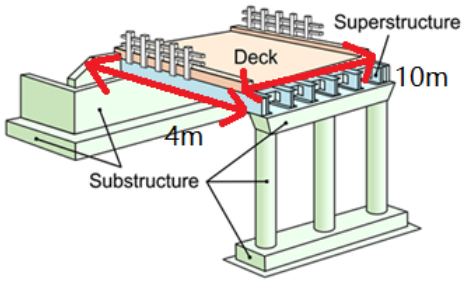


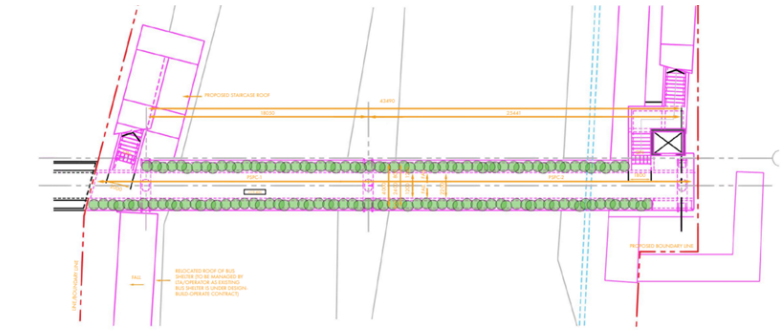
Figure 12: Steel Deck Framing dimensions [16]



Figure 13: Pedestrian walkway along the river [17]



Figure 14



Proposed Shrub Planting Scheme

Legend	Height (mm)	Spacing (mm)	Name of Shrub Species	Bushy	Flowering	Full Form
	400	300	*Ixora Super Pink*	✓	✓	✓
	200	250	*Phyllantus cochinchensis*	✓		✓
	600	300	*Jatropha spp*	✓		✓
	200	250	*Philodendron yellow*	✓		

Legend	Height (mm)	Spacing (mm)	Name of Shrub Species	Bushy	Flowering	Full Form
	400	250	*Pseudomussaenda flava*	✓		✓
	300	250	*Duranta 'Dark purple'*	✓		✓
	500	300	*Ficus Macrocarpa Gold*	✓		✓

Figure 15 [18]

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